

Advanced V&V for motor control

User Cse#3- Motor Control Sensitivity Analysis

Contents

- How does Advanced VV works?
- Structure for Advanced VV
- Harness Block.
- Scripts
- Report
- Folder structure.

Use Case #3

What is idea/aim behind running user case#2???

- To Change initial position of the motor.
- Starting motor at different initial positions.
- Covering max and min angle spam.
- Startup analysis.
- Should start motor at any initial position.
- Improve controller strategy.
- Test will be more of investigation.

Use Case #3

Tool: Transient Simulation Architecture

Motor Control: DTC for sBPM

Stream: Windy_Mcu_sBPM_Floating- **Version 03 and 04**

Board: WindyStripH50_230V

Motor: H27 Hefei [with fixed inductances]

Mechanical Model Variant: 1) Dyno Load

2) Mechanical Load(Washer)

Description: **Evaluate motor controllability under different initial angle of motor.(Thata_m)**

Ranges: Speed\torque matrix [to be applied in dyno load mode].

Speed [Drum]	Torque[Drum]		
50 rpm	0Nm	15Nm	25 Nm
500rpm	0Nm	5Nm	10Nm
1000 rpm	0Nm	3.5Nm	7Nm

Use Case #3

Initial Angle Theta_mechanical will be tested against one of the following parameter.

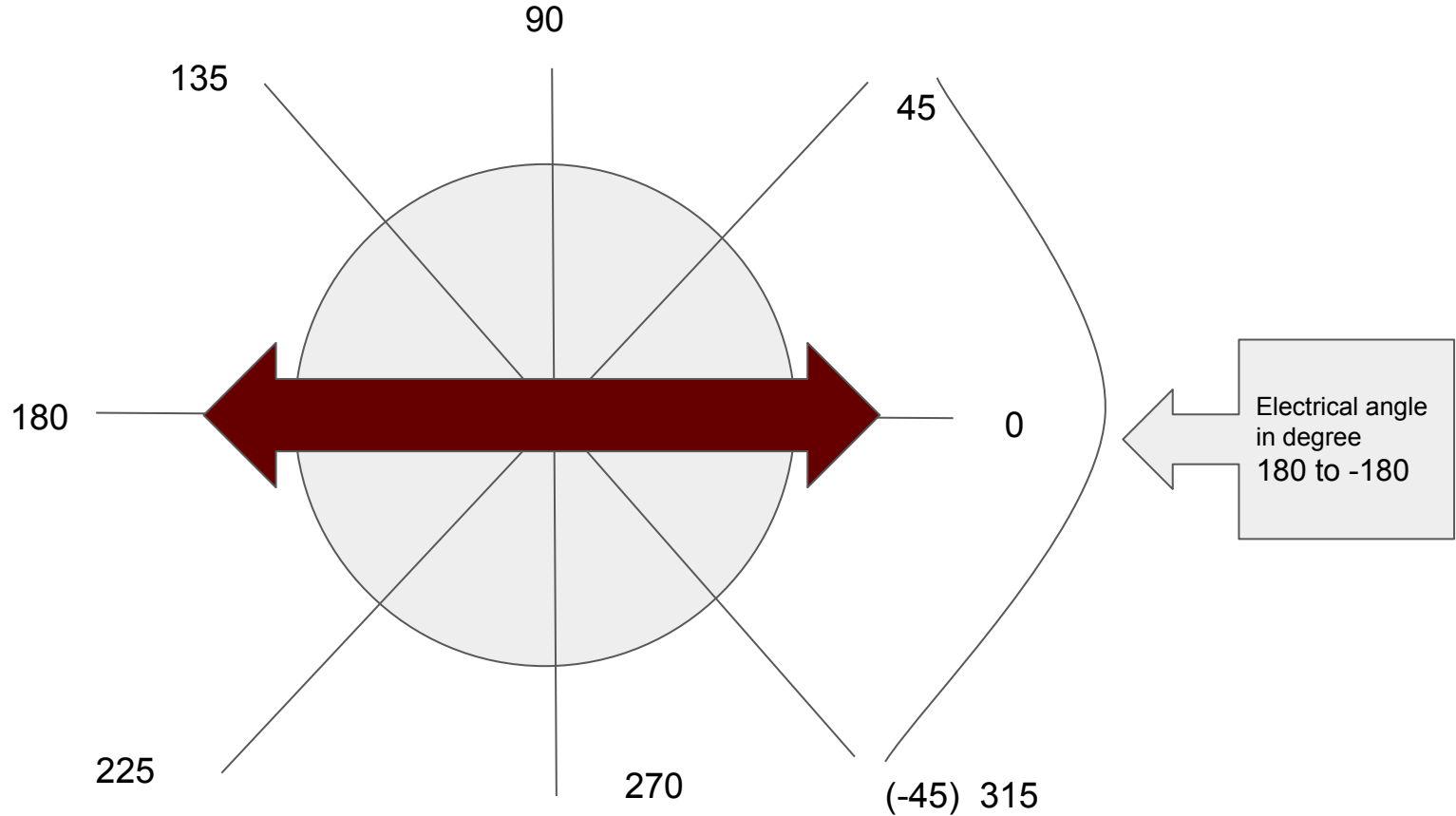
- Dyno Load.
- Acceleration.
- Ts (Resistance).
- JDrum.
- J Damp.

Only when Mechanical Load

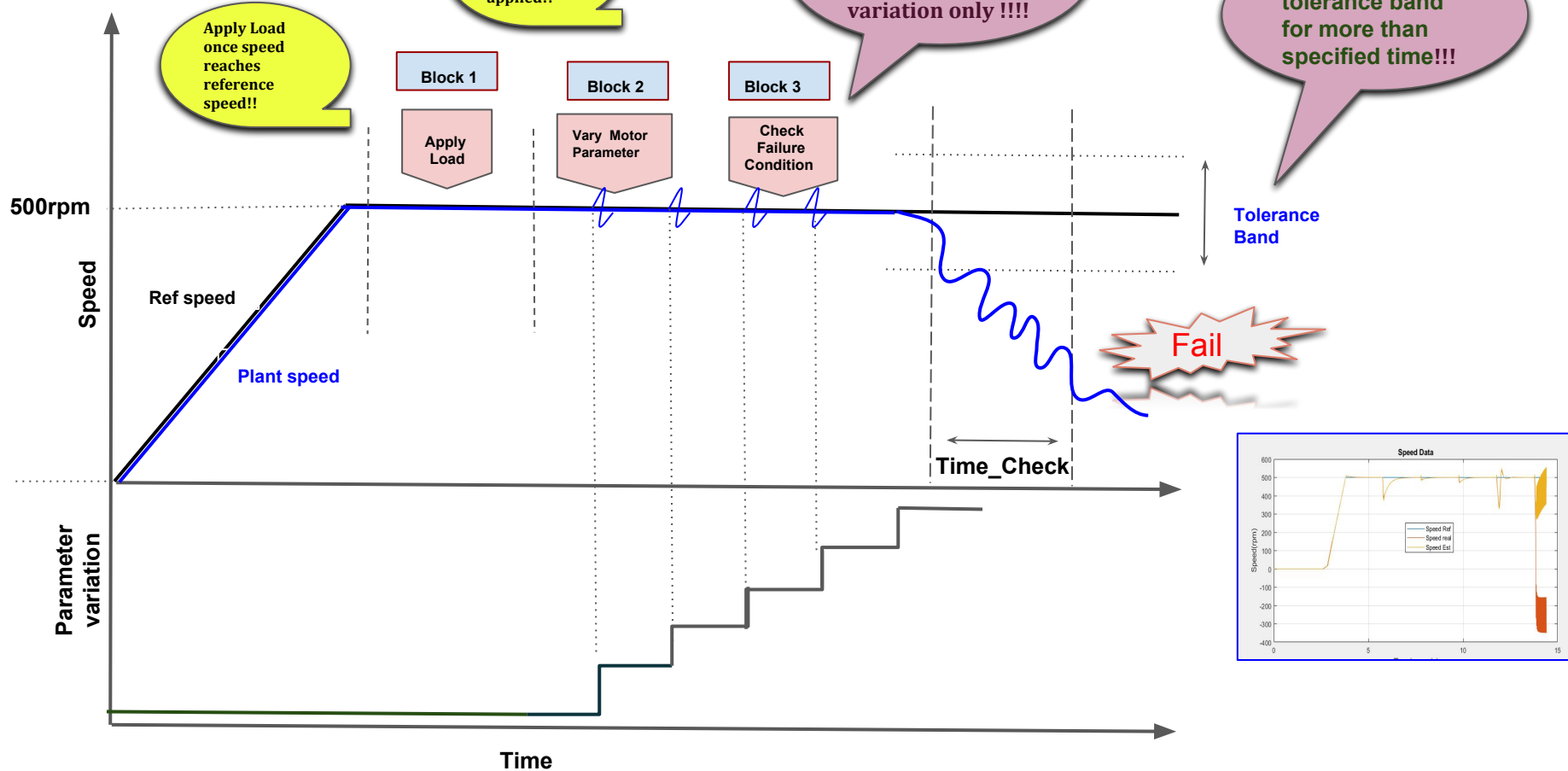
- The startup analysis need to be executed on
 - 2 source codes, with bypass of resistance estimation.
 - 2 Mechanical model variant.

Note- Case#3 will be carried out directly on Simulation Architecture.

Mechanical angle span in degree



Implementation



Structure

Advanced VV

Scripts

Automatic Script.m

--Allows to run the test with the given speed/torque.
--It manages the output.

Initialization.m

--Allows user to input the Parameters and limits.
--Allows the user for selection of type of load,stream and test.

Harness Model

Signal block

It Inputs the parameter variation signal to the module which is under test.

Exported block

Plant

+

Controller

In Case#3 Exported block is simulation architecture itself

Library

Library includes detailed documentation of the blocks created in Advanced VV.

Initialization Scripts

```
% Last Modified by v1.1 14-Feb-2019 16:08:20
%-----
% Author:      Madhura Patil
% Date:        October 12th, 2018
%-----

%-----
% Parameter Initialization
% Description: This Function is used to load the parameters variable given
% to simulink Model used for testing the sBPM by varying resistance and flux
% and thus interns temperature of stator and rotor temperature is varied.
%-----

%-----
% PLEASE NOTE -This script is used in Automatic_script_Temp.m file for calling parameters.
%-----

% Edit the below parameters to modify the response.
%-----

addpath(genpath(pwd));
%.....Input Parametr value.....%
Automatic_Test.Speed=[500;5000;1200]; % .....To enter Speed values(rpm).....%
Automatic_Test.Dyno_Load_Vector=[0.5;0.8;1.3]; % .....To enter Dyno Load values(Nm).....%
Automatic_Test.Acceleration=500; % .....To enter Acceleration of Speed(m/s).....%
Automatic_Test.Speed_Tolerance=20; % .....To enter Speed Tolerance in(%).....%
Automatic_Test.Percentage_Variation=50; % .....To enter the Percentage Variation of t

%...Flag setting for selection of variation of parameter.....%
%Uncomment either one of it as per the selection of test,that is by varying

%Automatic_Test.parameter_var= 'Flux';
%Automatic_Test.parameter_var= 'Rs';

%.....Demagnetisation value.....%
Automatic_Test.dm=-0.0021;

%.....Input Parametr value to Matlab Functions.....%

%.....Status Function.....%
Automatic_Test.Status_Time_Check=2; %...Status time check in (sec)
Automatic_Test.Status_Tolerance=0.05; %...Status Tolerance time in (s)

%.....Dyno Input Function.....%
Automatic_Test.Dyno_Step_Size=0.001; %...Dyno Step Size value in (s)

%.....Parameter Input Function.....%
Automatic_Test.Para_time_check=2; %...Generation of Prameter in (s)
```



Instruction to use m file



Enter/modify the motor parameters such as speed acc load etc



Enter/modify the controller parameters such as speed tolerance,% variation etc



Selection of test. For e.g by imposing noise on Vdc/labc.



Enter/modify other parameters as time checks for step size,failure time etc.

Automatic Scripts

To collect values of varied motor parameters at each step interval.

To plot the Report.

Naming the figure against its speed and torque.

Plot and format each figure by giving label,title,legends etc.

To save the Report and .fig file in created folder.

Function to calculate statistics (mean and standard deviation) of each signal at each interval.

Instructions

Open harness model and call initialization file.

Create folder to save figure and .csv file.

Masking parameters as per selection of file.

Create a Report format.

Create loop to run the test against Speed and Torque/Parameter matrix.

Simulate the Harness Model.

To collect statistics such as mean/standard deviation of Theta and Torque error.

```
%.....To collect the parameters values.....%
Speed_ref_prm=Automatic_Test.Speed_Ref*ones(size(Theta_Error_mean_values)); %..
Dyno_ref_prm = Automatic_Test.Dyno_Load * ones(size(Theta_Error_mean_values)); %..

%.....To get the data of parameter variation....%
if (strcmp(Automatic_Test.parameter_var,'Rs'))
step = Automatic_Test.Percentage_Variation*Motor.Electrical.Rs_a_Ref/100;
Parameter_Var = (Motor.Electrical.Rs_a_Ref:step:(Motor.Electrical.Rs_a_Ref+step*(1e

elseif(strcmp(Automatic_Test.parameter_var,'Flux'))

step = Automatic_Test.Percentage_Variation*Motor.Electrical.PM_Flux_f01/100;
Parameter_Var = (Motor.Electrical.PM_Flux_f01:step:(Motor.Electrical.PM_Flux_f01+st
end
```

```
%.....To update the results table.....%
Results = [Results ; table(Speed_ref_prm,Dyno_ref_prm,Parameter_Va
Theta_Error_std_values',Torque_Error_mean_values',Torque_Error_
Speed_Plant_mean_values',Speed_Plant_std_values', ...
'VariableNames',{'Speed_ref','Torque_ref', Automatic_Test.parameter
'Theta_error_std', 'Torque_error_mean', 'Torque_error_std', 'Sp

%.....Naming the figure file against speed and torque....%
r=strcat('Speed_',num2str(Automatic_Test.Speed(i)), '_Load_',strrep(n

%..... To Plot the Speed, torque and theta error.....%
y = figure('Name','Run','NumberTitle','off');
subplot(2,2,1);
plot(Speed);
ylabel('Speed(rpm)')
title('Speed Data')
legend('Speed Ref','Speed real','Speed Est')
grid on;
subplot(2,2,2);
```

```
function [interval_index] = get_intervals(parameter_signal)
parameter_signal_shifted = [parameter_signal(1) parameter_signal];
parameter_signal = [parameter_signal parameter_signal(end)];
index_signal = parameter_signal_shifted - parameter_signal;

k=find(index_signal==0);
if k
%....Adding first interval with nominal value....%
interval_index = [k(1)-(k(2)-k(1)) k];
else
interval_index = [];
end
```

```
function [mean_values,std_values] = statistics(signal, interval_index)
n=length(interval_index)-1;
mean_values = zeros(1,n);
std_values = zeros(1,n);
for i=1:n
interval=interval_index(i+1)-interval_index(i); % To get the interv

start_1=interval_index(i)+floor(interval*0.2); % To get the data at
final=interval_index(i+1)-floor(interval*0.01); % To get the data at

windowed_value=signal(start_1:final);
mean_values(i)=mean(windowed_value);
std_values(i)=std(windowed_value);
end
```



```
% Last Modified by v1.1 14-Feb-2019 16:08:20
% Author: Madhura Patil
% Date: October 12th, 2018
%-----
% Description: This Script is used to run and test performance of motor
% against defined speed and load torque.
% It checks the robustness of the motor under test by generating the failure
% values of the tests.
%-----
% Requirements to run the test
% Electrical Requirement:- Permanent Magnet Motor
% Mechanical Requirement:- Dyno Load
%-----
% Please Note- Before running this script, Parameter Initialization file has
% to be updated to edit the parameters required to run the test
%-----

%.....Opening the model model.....%
addpath('..\..\..\Advanced_V_V\tests\parameter_variation\harness_model')
open('Harness_Temp_Vary.slx');

Parameter_Initialization;

%.....To create and name the folder by current date and time.....%
dname = uigetdir('C:\N\');
currDate = strrep(datestr(dacetime), ' ', '_');
basestrcat(dname, '\',currDate);
delete that.
%.....Parameter Initialization for the selection of Ts and Tr signalInput..
h1 = Simulink.findBlocks(gcs,'Name','Parameter Value');
h2 = Simulink.findBlocks(gcs,'Name','Temp Input');
%.....Parameter Initialization for the selection of Parameter (RS) signalIn
if (strcmp(Automatic_Test.parameter_var,'Rs'))
set_param(h1,'init_value',Motor.Electrical.Rs_a_Ref);
set_param(h2,'init_value',Motor.Electrical.Rs_a_Ref);
set_param(h2,'Temp_Ref',Motor.Mechanical.Rs_Temperature_Ref);
set_param(h2,'alpha',Motor.Electrical.alpha_s');
elseif(strcmp(Automatic_Test.parameter_var,'Flux'))
%.....Parameter Initialization for the selection of Parameter (Flux) signal
set_param(h1,'init_value',Motor.Electrical.PM_Flux_f01);
set_param(h2,'init_value',Motor.Electrical.PM_Flux_f01);
set_param(h2,'Temp_Ref',Motor.Mechanical.PM_Temperature_Ref);
set_param(h2,'alpha',Automatic_Test.dn1);
else
msg='Please select type of variation either Rs or Flux in Automatic_Test.p
error(msg)
end

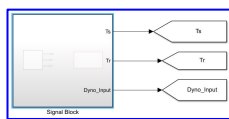
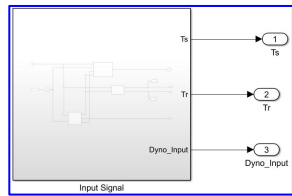
%.....Creates an empty table to collect results....%
Results = table(1,[],1,[],1,[],1,[],1,[],1,[],1,[],1,[],1,[],1,[],1,...
'VariableNames',{'Speed_ref','Torque_ref', Automatic_Test.parameter_var,'Th
'Theta_error_std', 'Torque_error_mean', 'Torque_error_std', 'Speed_mean'

%.....Iteration.....%

for i= 1:length(Automatic_Test.Speed)
Automatic_Test.Speed_Ref=Automatic_Test.Speed(i);
for j=1:length(Automatic_Test.Dyno_Load_Vector)
Automatic_Test.Dyno_Load=(Automatic_Test.Dyno_Load_Vector(j));

%....Simulate the Model.....%
sim('Harness_Temp_Vary.slx');

%..... To collect statistics.....%
if (strcmp(Automatic_Test.parameter_var,'Rs'))
interval_index = get_intervals(Ts.signals.values');
elseif(strcmp(Automatic_Test.parameter_var,'Flux'))
```

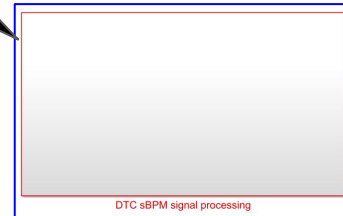


Input Signal Block

Signals Input

Signals and Scope

Signal Processing scopes



Block-1
Speed
Dyno_Load

Block-2
Parameter
Variation

Block-3
Condition check

Speed
Accele

Speed Input

Speed attained

Dyno Status

Dyno Input

Dyno Load Input

Step Signal

Noise Input

Vdc

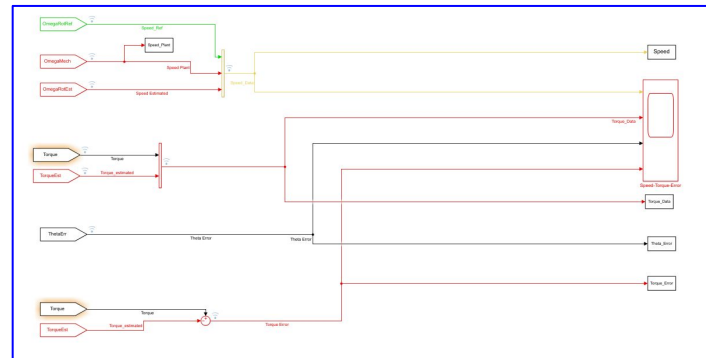
Iabc

Dyno status
Speed

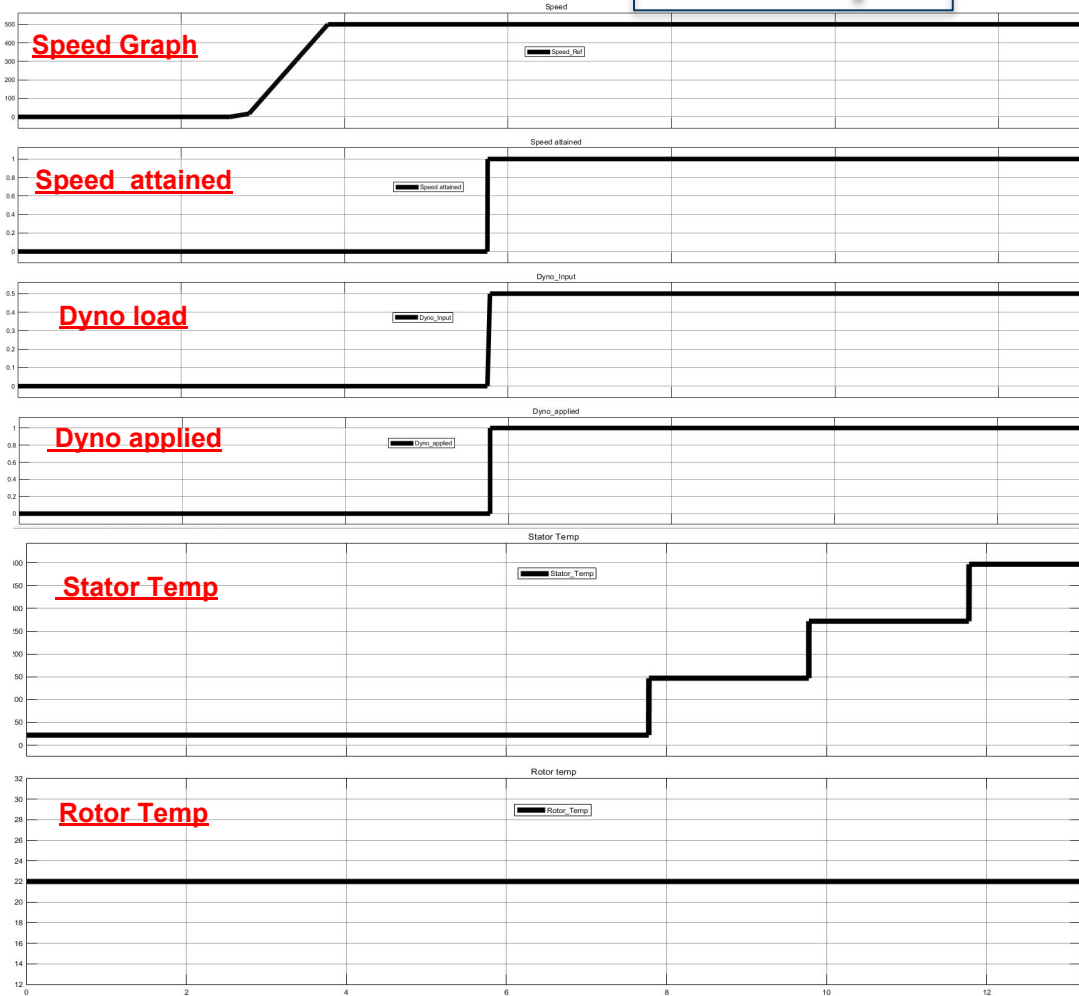
Condition Check

If speed is out of tolerance bound for more than specified time

Stop



Sinal Graphs



Speed Graph



When plant speed reaches ref speed, **Speed attained** Status is high.



Speed attained Status is high **Dyno load** has applied.



When Dyno load has applied, **Dyno applied** status is high.



Once Dyno applied status is high, parameter variation begins.

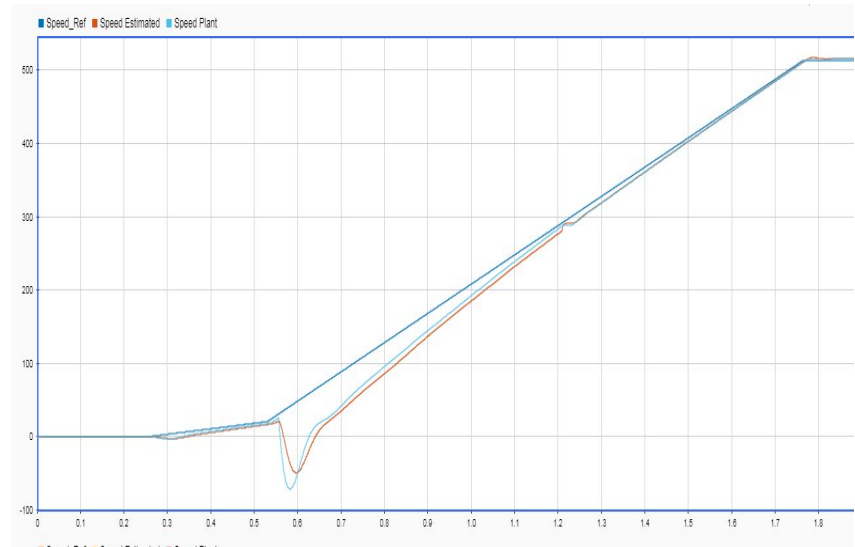
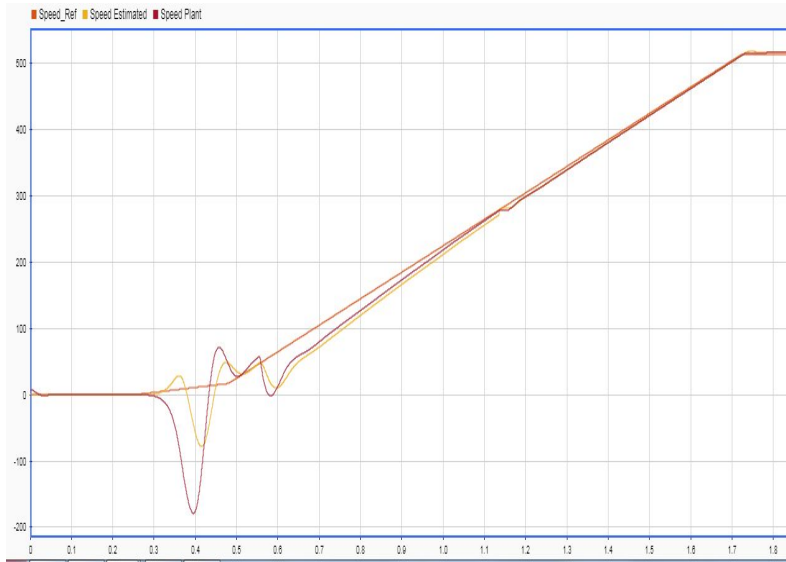


Other parameter retains its nominal value during the test.

Types of start up at fail condition

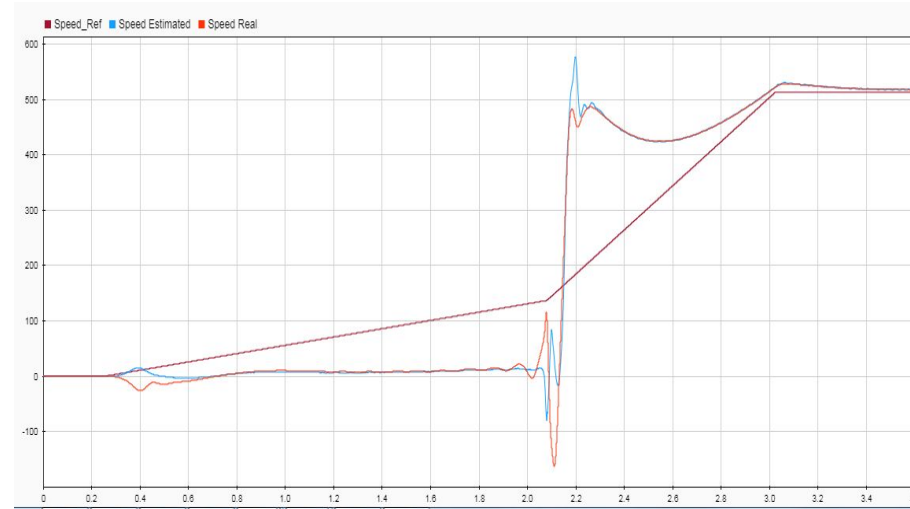
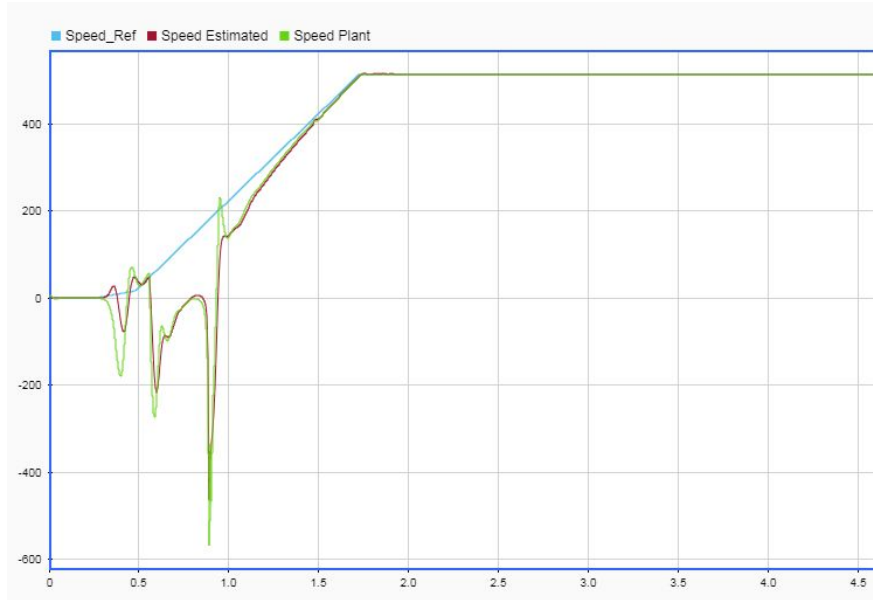
Bad Startup

- During Start up, speed has encountered peaks for a short duration.
- Time duration is configurable through Para_time_check_1. In Para_Intia .m file.
- Speed tolerance is configurable through Speed_tolerance value In Para_Intia .m file.



Worse Startup

- During start up, speed has encountered more than one peaks of a short duration. Also speed is not upto mark.
- Time duration is configurable through Para_time_check_1. In Para_Intia .m file.
- Speed tolerance is configurable through Speed_tolerance value in Para_Intia .m file.



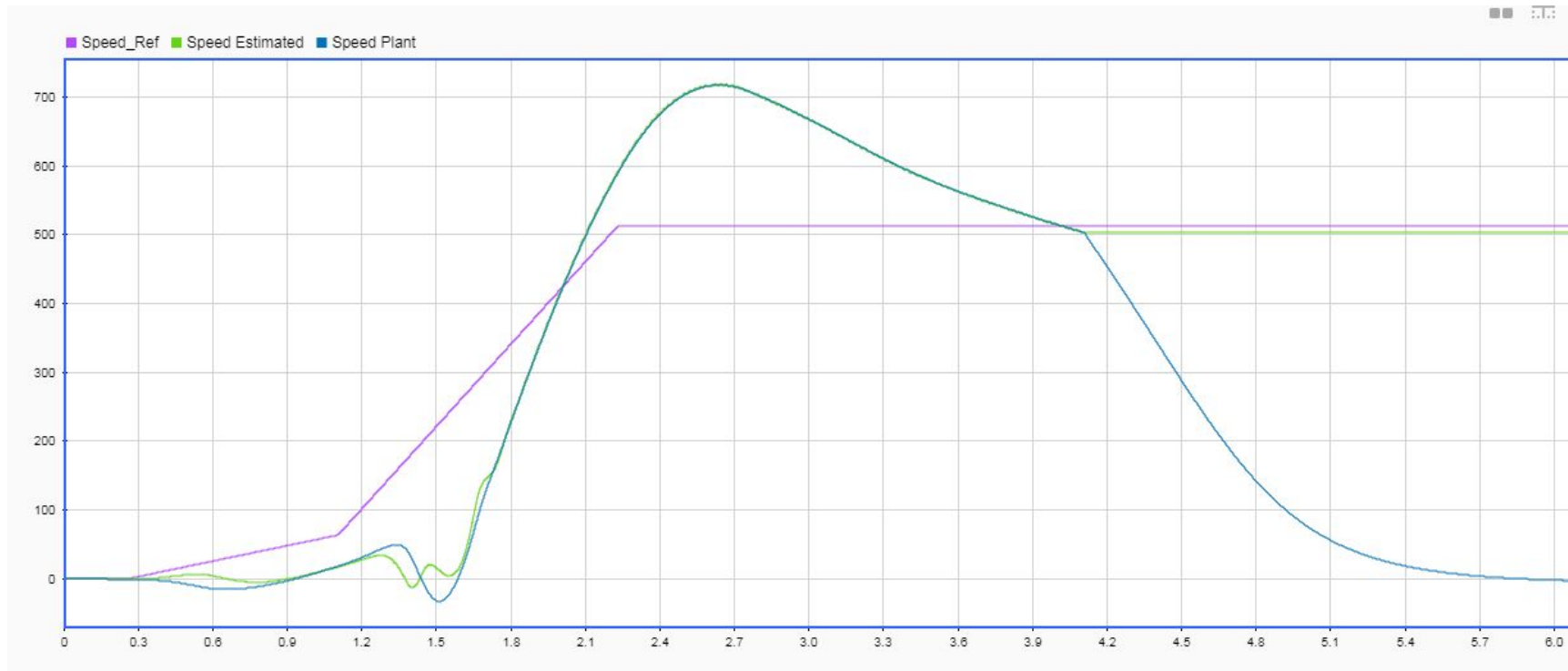
Failure Condition_1

- This Condition looks into the controller capability such that speed of plant is aligned with Ref speed.
- If speed is failed to reach the reference speed and further goes out of bound.

Condition

- When the speed is beyond the speed tolerance value for more than defined time, than it passes Fail_1 condition.
- Time duration is configurable through Para_time _check_2.
- Speed tolerance is configurable through Speed_tolerance value.

Failure Condition_1



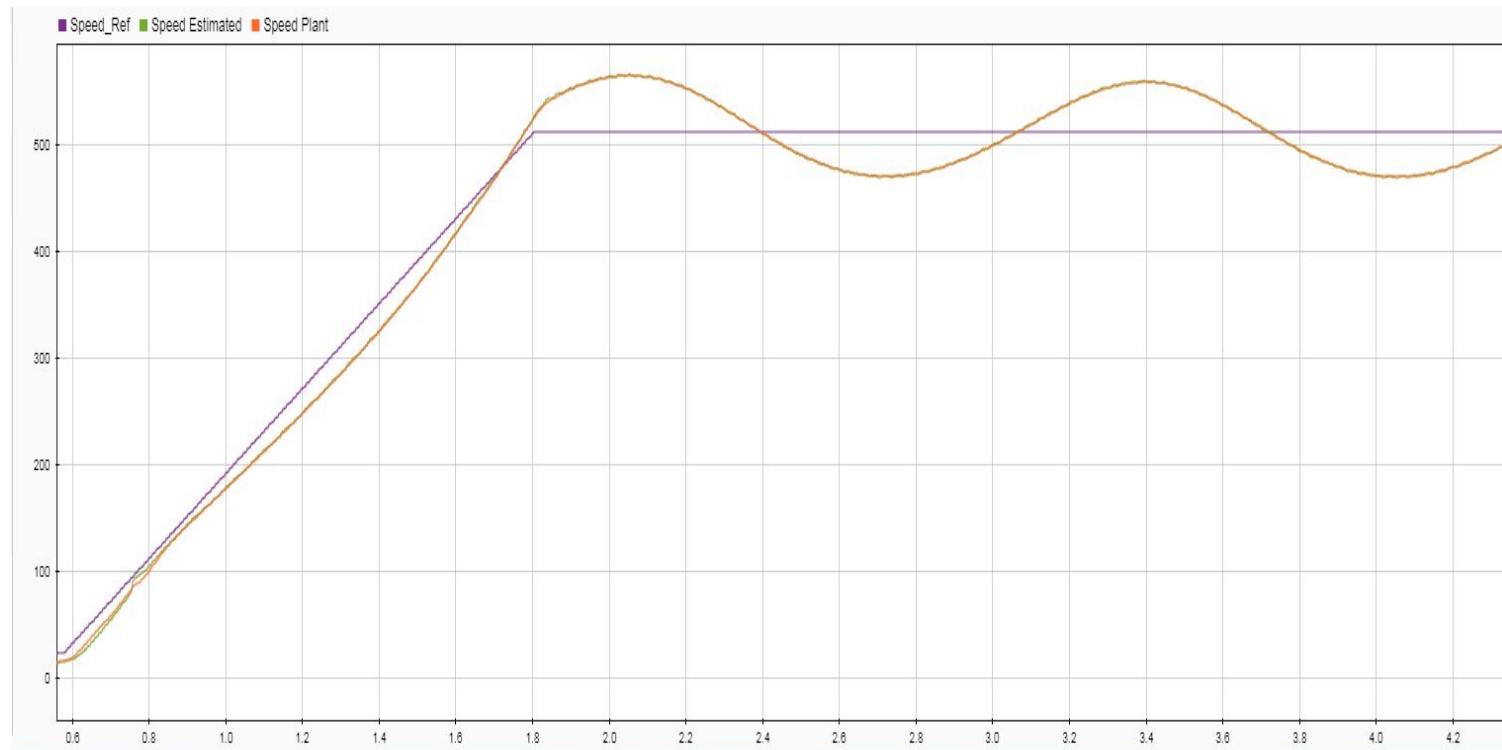
Failure Condition_2

- If speed is failed to reach the reference speed and has oscillatory response and never reaches its defined value.

Condition

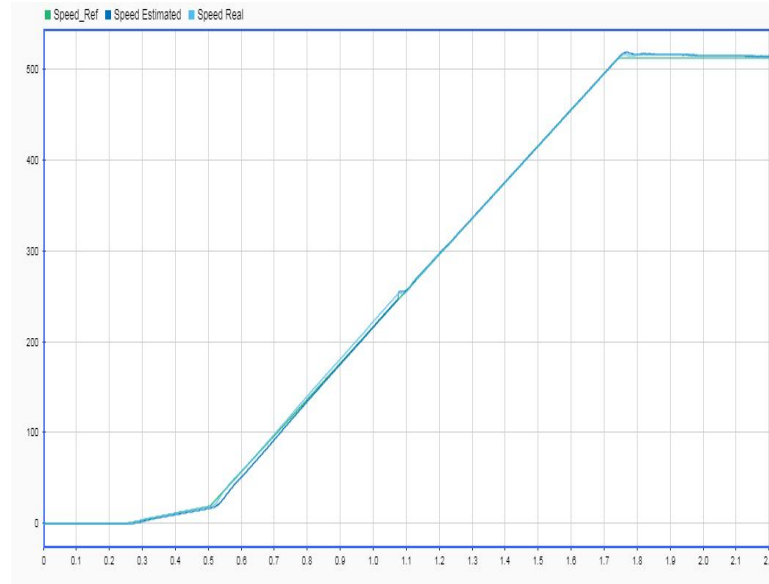
- Once the speed has reached the speed ref and if speed is greater than defined tolerance it passes Fail_2 condition.
- Time duration is configurable through Para_time _check_3.
- Speed tolerance is configurable through Speed_tolerance value_1.

Failure Condition 2

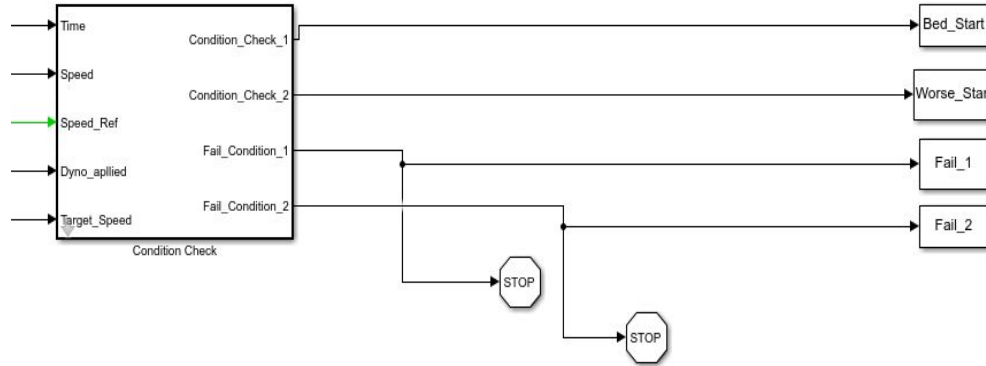


Pass Condition

- During startup and also once the speed reaches defined reference speed, there are no abnormalities encountered.
- Speed is attained without much deviation from its normal speed.
- However small jerks and uncertainties having low level has been neglected.



Condition Block



Difference between Bad and Worse Start up?

----- if (Tolerance > time_check)

Count+1

if(Speed ref>= Target speed)

If Count>1

Worse startup

Elseif (Count=1)

Bad startup

Report Generation

- Version
- Mechanical model variant.
- Max/Min Electrical angle position.
- Controller parameter.
- Status
 - Bed Start up
 - Worse Start up
 - Fail_1
 - Fail_2

Stream_Ver	Angle	Jdrum	Status
04_Mech	-180	4.8	Fail_1
04_Mech	-150	4.8	Fail_1
04_Mech	-120	4.8	Fail_1
04_Mech	-90	4.8	Fail_1
04_Mech	-60	4.8	Fail_1
04_Mech	-30	4.8	Fail_1
04_Mech	0	4.8	Fail_1
04_Mech	30	4.8	Fail_1
04_Mech	60	4.8	Fail_1
04_Mech	90	4.8	Fail_1
04_Mech	120	4.8	Fail_1
04_Mech	150	4.8	Fail_1
04_Mech	180	4.8	Fail_1

Report-Investigation of failure value

	Nominal Values	Limit	Version 03	Version 04	Load Type
Res	3	Max	----	9.336	Dyno
Res	3	Min	----	1.96	
Dyno			----		
Res	3	Max	7.53	8.736	Mech
Res	3	Min	2.85	1.96	
JDrum	0.063	Max	3.5	4.8	Mech
		Min	0.063	0.063	
JDamp	0.28	Max	0.5	1	
		Min	0.28	0.28	

Folder structure

MCU_Sim_Arch

advanced_V_V

lib (Library)

This folder contains library of each matlab blocks created for advanced VV testing purpose.

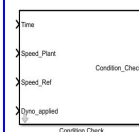


Go to library browser

Condition Check_Temp

Description

This Matlab function block is used to keep check on failure of the test.



Input

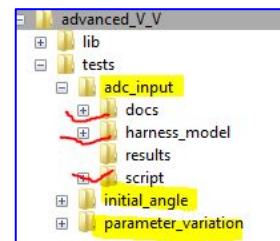
1. Time is real simulation time which is used to calculate failure conditions.
2. Speed_Plant is the output speed from the plant.
3. Speed_Ref is the given reference speed by the user.
4. Dyno_Applied is the status to know whether Dyno load is applied or not.

Output

The output Condition Check will send the signal to stop the simulation if failure condition is true

tests

--adc_input
--initial-Angle
--Parameter variation



Docs-Help document/guidelines to conduct tes.

Harness_model- Contains harness model as .slx file.

Results - Folder to save the results.

Script - Contains two script, Automatic and Initialization.m files